APS Algorithmic Problem Solving-II
Experiment Answers (1–10)

Experiment 1: Pyramid Star Pattern

A pyramid star pattern can be printed using nested loops where the outer loop controls rows and inner loops handle spaces and stars. The number of stars follows (2*i - 1).

```
public class Pyramid {
    public static void main(String[] args) {
        int n = 5;
        for(int i=1; i<=n; i++) {
            for(int j=1; j<=n-i; j++)
                System.out.print(" ");
            for(int k=1; k<=2*i-1; k++)
                System.out.print("*");
            System.out.println();
        }
    }
}
```

Experiment 2: Number Triangle Pattern

This program prints numbers in a triangular format using nested loops where each row displays numbers from 1 to the row number.

```
public class NumberTriangle {
    public static void main(String[] args) {
        int n = 5;
        for(int i=1; i<=n; i++) {
            for(int j=1; j<=i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

Experiment 3: One-Dimensional Array

This program demonstrates declaration, initialization, and display of elements of a 1D array.

```
public class ArrayDemo {
    public static void main(String[] args) {
        int arr[] = {10, 20, 30, 40, 50};
        for(int i=0; i<arr.length; i++) {
            System.out.println(arr[i]);
        }
    }
}
```

Experiment 4: Largest and Smallest in Array

This program finds the maximum and minimum values in a 1D array by iterating and comparing elements.

```
public class MaxMin {
    public static void main(String[] args) {
        int arr[] = {10, 5, 20, 8, 25};
        int max = arr[0], min = arr[0];

        for(int i=1; i<arr.length; i++) {
            if(arr[i] > max)
                max = arr[i];
            if(arr[i] < min)
```

```
                min = arr[i];
        }

        System.out.println("Max: " + max);
        System.out.println("Min: " + min);
    }
}
```

Experiment 5: 3x3 Matrix Display

This program creates and displays a 3x3 matrix using a 2D array.

```
public class MatrixDemo {
    public static void main(String[] args) {
        int arr[][] = {
            {1,2,3},
            {4,5,6},
            {7,8,9}
        };

        for(int i=0; i<3; i++) {
            for(int j=0; j<3; j++) {
                System.out.print(arr[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Experiment 6: Multiplication Table (2D Array)

This program generates a multiplication table using a 2D array.

```
public class MultiplicationTable {
    public static void main(String[] args) {
        int table[][] = new int[5][5];

        for(int i=0; i<5; i++) {
            for(int j=0; j<5; j++) {
                table[i][j] = (i+1)*(j+1);
                System.out.print(table[i][j] + "\t");
            }
            System.out.println();
        }
    }
}
```

Experiment 7: Sum of 2D Array

This program calculates the sum of all elements in a 2D array.

```
public class Sum2D {
    public static void main(String[] args) {
        int arr[][] = {
            {1,2,3},
            {4,5,6},
            {7,8,9}
        };

        int sum = 0;

        for(int i=0; i<3; i++) {
            for(int j=0; j<3; j++) {
                sum += arr[i][j];
            }
        }

        System.out.println("Sum: " + sum);
    }
}
```

Experiment 8: String Operations

This program counts vowels in a string and replaces a word using built-in string methods.

```
public class StringOps {
    public static void main(String[] args) {
        String str = "Hello World";
        int count = 0;

        for(int i=0; i<str.length(); i++) {
            char ch = str.charAt(i);
            if("aeiouAEIOU".indexOf(ch) != -1)
                count++;
        }

        System.out.println("Vowels: " + count);

        String newStr = str.replace("World", "Java");
        System.out.println("Updated String: " + newStr);
    }
}
```

Experiment 9: User-Defined Methods

This program uses methods to calculate the area of a circle and find the maximum of three numbers.

```
public class MethodsDemo {

    static double area(double r) {
        return Math.PI * r * r;
    }

    static int max(int a, int b, int c) {
        return Math.max(a, Math.max(b, c));
    }

    public static void main(String[] args) {
        System.out.println("Area: " + area(5));
        System.out.println("Max: " + max(10, 20, 15));
    }
}
```

Experiment 10: Recursion (Factorial & Fibonacci)

This program demonstrates recursion by calculating factorial and Fibonacci numbers.

```
public class RecursionDemo {

    static int fact(int n) {
        if(n == 0)
            return 1;
        else
            return n * fact(n-1);
    }

    static int fib(int n) {
        if(n <= 1)
            return n;
        else
            return fib(n-1) + fib(n-2);
    }

    public static void main(String[] args) {
        System.out.println("Factorial: " + fact(5));
        System.out.println("Fibonacci: " + fib(6));
    }
}
```